

# Life After NGINX Ingress Controller

Architecting the exit - and a low-downtime live migration

# NGINX Ingress Controller Is Dead, and You Are Probably Running It

24 Mar 2026

The repository was archived and the solution is deprecated.  
Meaning: no releases, no bug fixes, no CVE patches.

~50%

of Kubernetes clusters still run it as their ingress data plane.

8 CVEs

already accumulated, led by CVE-2026-42945 - a critical heap overflow ("NGINX Rift") rated 9.2 under CVSS v4.0. Permanently unpatched.

SPEAKER

# Aviv Kabesa

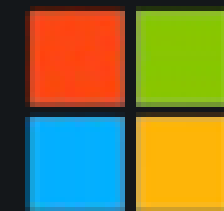
Cloud Solutions Architect, Microsoft

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Previously Viz.AI

DevOps Group Lead at TechGym

Love trekking



Microsoft



# What Exactly Died, and What Did Not

|                                      |                                        |            |                                         |
|--------------------------------------|----------------------------------------|------------|-----------------------------------------|
| NGINX Ingress Controller (community) | kubernetes/ingress-nginx - SIG Network | DEPRECATED | free · Apache 2.0 no support to buy     |
| NGINX Ingress Controller             | nginx/kubernetes-ingress - F5          | ACTIVE     | free on NGINX OSS<br>paid on NGINX Plus |
| NGINX Gateway Fabric                 | F5 - Gateway API implementation        | ACTIVE     | free on NGINX OSS<br>paid on NGINX Plus |
| NGINX, the web server                | nginx.org - unrelated release train    | UNAFFECTED | free on NGINX OSS<br>paid on NGINX Plus |

# An Architecture Problem, Not a Staffing Problem

## FAULT 01

### Reload storms

Any config change regenerates the whole `nginx.conf` from a Go template and reloads. In-flight connections drop; busy clusters reload constantly.

## FAULT 02

### Snippet injection

Any namespace could inject raw NGINX config into a shared process through `configuration-snippet`. Disabled by default since v1.9 - still enabled in plenty of clusters. Unmaintainable, and an attack surface.

## FAULT 03

### Unauthenticated webhook

Validation ran `nginx -t` on untrusted input with no auth. Network reachability equalled code execution - the root of CVE-2025-1974.



# Is This Actually Your Emergency?

Answer two questions before you plan anything

- 01 Is the ingress internet-facing, or internal to a VPC?
- 02 Is the admission webhook network-reachable from untrusted pods?

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Two NO's might mean you have time.  
The EOL is real; the urgency is not evenly distributed.

## The toxic combination

Risk is real where exposure + vulnerability + access to sensitive data all meet. Absent that overlap, plan properly instead of rushing.

## The log4j lesson

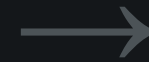
Critical to fix - but not for every service, on day one. Spending all your effort on one loud threat is how the quiet one gets in.

# Swap First, Gateway API Later

1  
NOW

## Swap the controller

Same Ingress objects. New binary behind them.



2  
WHEN READY

## Adopt Gateway API

Per route, on your own schedule.

# The Shortlist



**Traefik**



**Envoy Gateway**



**Istio**



**Cilium Gateway**



**Cloud controllers**



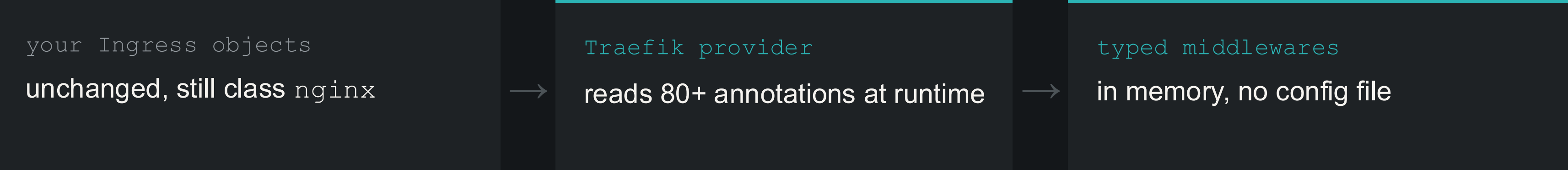
**NGINX OSS**



# Traefik v3.7

both APIs, one instance  
free · MIT

Nothing in your cluster changes shape



0

manifest rewrites to swap

0

reloads - Go router, no NGINX binary

snippets

not translated - reported, not applied

# Envoy Gateway

Gateway API only **free** ·  
Apache 2.0 · CNCF

`file reload`   `render config` → `reload process` → `drain connections`

`xDS`   `control plane` → `gRPC stream` → `live proxy fleet`

provisioning

a Gateway spawns its own Envoy  
Deployment + Service

policy

rate limit, OIDC, JWT, retries - as  
CRDs

the trade

no Ingress API - every manifest  
rewritten first

# Istio and Cilium

free · vendor / enterprise builds paid

## Istio mesh optional

Gateway resource

provisions a dedicated Envoy deployment

ambient (optional)

ztunnel = L4 + mTLS · waypoint = L7 where attached

## Cilium built into the CNI

L4

eBPF, in-kernel - no proxy hope

L7

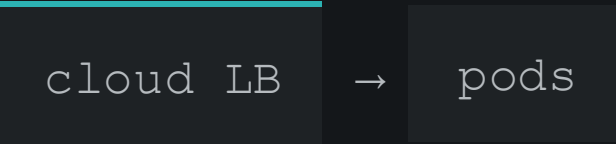
Envoy, per-node DaemonSet - no extra controller

Worth it only if you already run them. Neither reads NGINX annotations.

# Cloud Controllers and F5 NGINX

the two people forget

## Cloud controllers



data plane lives outside the cluster

## F5 NGINX



different codebase - still a full migration

Nothing to patch · single-cloud  
controller free · LBs billed hourly

Keeps annotations + NGINX templating  
free on OSS · paid on Plus

# Six Options, Six Starting Points

## Traefik

both APIs · MIT

free · Traefik Hub is paid

Pure Go, no NGINX binary, no reloads.  
v3.7 reads 85+ NGINX annotations  
natively. Lowest-friction swap.

## Envoy Gateway

Gateway API only

free · Apache 2.0, CNCF

Reference implementation on Envoy xDS.  
Declarative rate limiting, OIDC, AI  
Gateway extensions.

## Istio

only if you run the mesh

free · vendor builds paid

Replaces the NGINX Ingress Controller  
with zero new components. Ambient:  
ztunnel L4/mTLS, waypoints for L7.

## Cilium Gateway

only if it is your CNI

free · Isovalent Ent. paid

Built in when Cilium is installed. eBPF L4  
in-kernel, Envoy for L7 in userspace.

## Cloud controllers

single-cloud only

controller free, LBs billed  
hourly

AWS LBC, Azure AGC and GKE Gateway.  
Real cloud LBs, not in-cluster proxies.

## NGINX OSS

F5, Ingress API only

free · NGINX Plus paid

Different codebase - still a migration.  
Keeps you on annotations, and on  
NGINX's templating model.



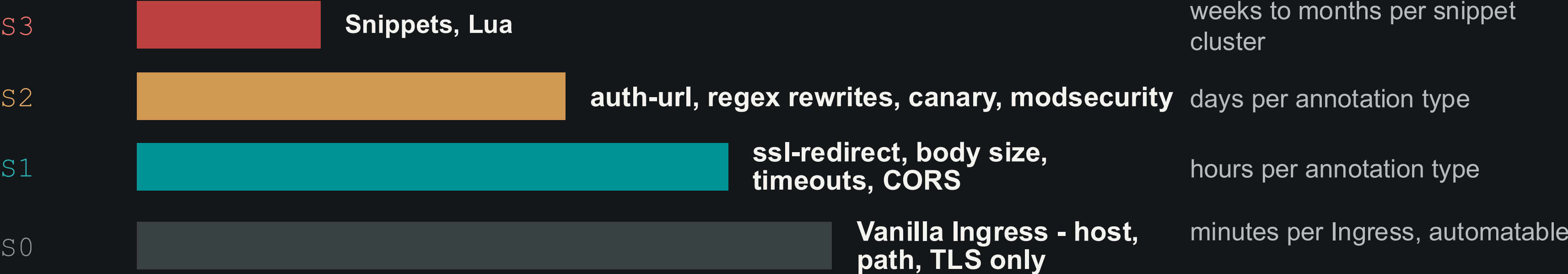
# Comparison Matrix

|                   | Traefik v3.7     | Envoy Gateway     | Istio              | Cilium GW        | AWS LBC          |
|-------------------|------------------|-------------------|--------------------|------------------|------------------|
| Ingress API       | Yes              | No                | Yes                | Yes              | Yes              |
| NGINX annotations | 85+ native       | None              | None               | None             | None             |
| Architecture      | Pure Go, dynamic | Envoy / xDS       | ztunnel / waypoint | eBPF / Envoy     | Cloud LB         |
| Reload problem    | None             | None (xDS)        | None               | None (eBPF)      | N/A              |
| L7 rate limiting  | Middlewares      | Yes, token cost   | Yes                | Basic            | Cloud-managed    |
| Complexity        | Low              | Medium            | High               | Medium-high      | Low              |
| Maturity          | Since 2016       | Moderate          | Very               | Moderate         | GA 2026          |
| Cost              | Free (Hub paid)  | Free              | Free               | Free (Ent. paid) | LB billed hourly |
| Best for          | Drop-in swap     | Greenfield GW API | Existing mesh      | Existing CNI     | Single-cloud     |

# Choosing: First YES Wins



# Count Annotations, Not Ingresses



The NGINX Ingress Controller ships 115+ annotations; most teams use 10–30. Audit first - 70%+ of "unsupported" ones are usually safe to drop.



# The Universal Migration Pattern

|    |                            |                                                                                                                                                                                                                                                                                |
|----|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01 | <b>Map first</b>           | List every NGINX annotation by frequency, find all snippet usage, categorise S0-S3. The estimate comes from this, nothing else.                                                                                                                                                |
| 02 | <b>Deploy side-by-side</b> | New controller on its own <code>IngressClass</code> , no production traffic. Set <code>ingressClassName</code> explicitly on every Ingress.                                                                                                                                    |
| 03 | <b>Canary DNS shift</b>    | Move DNS or the cloud LoadBalancer to the new controller. Weighted records if your provider supports them; lower the TTL a day ahead.                                                                                                                                          |
| 04 | <b>Bake, then clean up</b> | Keep the NGINX Ingress Controller running 24-48h after the DNS switch for stale resolvers. Delete the webhook first. Annotate the <code>IngressClass</code> <code>helm.sh/resource-policy: keep</code> or Helm takes it - and every Ingress referencing it - with the release. |

LIVE DEMO

# Live Migration to Traefik on AKS

An AKS cluster running the NGINX Ingress Controller: audit the annotations, deploy side-by-side, put a catch-all in front, then peel routes off one at a time.

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01 - map

02 - side-by-side

03 - catch-all swing

04 - Gateway API

[github.com/avivkabesa/cncf-nginx-migration-demo](https://github.com/avivkabesa/cncf-nginx-migration-demo)

# What to Do Tuesday Morning

01 - Run the audit. Know what you depend on.

```
kubect1 get ingress -A -o json | jq -r '
.items[].metadata.annotations // {}
| to_entries[]
| select(.key | startswith(
    "nginx.ingress.kubernetes.io/"))
| .key' | sort | uniq -c | sort -rn
```

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## 02 - Assess your exposure

Internet-facing? Secrets in reach? Webhook exposed? Yes to all three: move now. Otherwise plan it properly.

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## 03 - Pick your path

Already on Cilium or Istio: use what you have. Heavy annotations: Traefik. Greenfield: Envoy Gateway.

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## Then migrate one Ingress

The simplest, lowest-risk route. Work up the pyramid from there - do not boil the ocean.

## TAKEAWAYS

- 01 The repository is archived and the patch path is gone - but urgency is a function of your exposure, not the date.
- 02 Your annotations, not your Ingress count, decide whether this is an afternoon or a quarter.
- 03 Swap the controller now, adopt Gateway API on your own schedule.

# Thank you

Everything from the demo is in the repo -  
Helm values, the catch-all route, and the  
Gateway API manifests.

Aviv Kabesa



demo repo  
`github.com/avivka`



LinkedIn  
`in/aviv-kabesa`